

DP exponents to one loop — Gunnar’s calculation, side-by-side with CFAC
 Short note. Reply to Gunnar’s 1 May 2026 letter.
 (Long version with all algebra: [reply.tex](#); full mechanisation: [paper/cfac/gribov.tex](#).)

The claim, in one sentence

CFAC reproduces every counting factor that enters the 1-loop DP RG — the diagram inventory, the symmetry factors, the IBP relation, the four Z -channel coefficients — by formal manipulation of the vertex polynomial $\phi(G) = 1 + G^2$, without enumerating diagrams or solving any integral except one.
The RG itself still has to be done. CFAC simply hands you all the numbers you need to do it.

This note shows Gunnar’s lines on the left and CFAC’s parallel derivation on the right. The numbers match line by line. We never substitute a result we did not derive; the only analytical input is the value of one Schwinger integral, $\mathcal{I}_{-O-}$, and that is admitted up front.

1. What is taken as given

Taken as given (the inputs)

The reader is assumed to know, and we do not re-derive:

1. **The DP action and free propagator.** $\mathcal{S} \supset g(\tilde{\phi}\phi^2 - \tilde{\phi}^2\phi)$, free propagator $(-i\omega + Dk^2 + r)^{-1}$, causal-time / Reggeon convention.
2. **The RG framework.** Multiplicative renormalisation, Z -factors, $\beta_{\bar{g}} = \mu\partial_\mu\bar{g}$, $\gamma_X = \mu\partial_\mu \ln Z_X$, dimensional regularisation in $d = 4 - \varepsilon$.
3. **The exponent dictionary.** $\eta = -2\gamma_\phi - \gamma_D$, $z = 2 + \gamma_D$, $1/\nu = 2 - (\gamma_r - \gamma_D)$.
4. **The Reggeon Ward identity.** $Z_\phi = Z_{\tilde{\phi}}$.
5. **One Schwinger integral.**

$$\mathcal{I}_{-O-}(\omega, k, r) = A \Gamma(\varepsilon/2) \frac{(-i\omega + Dk^2/2 + 2r)^{d/2-1}}{1 - d/2}, \quad A \equiv K_d/D^{d/2}.$$

This is the textbook bubble in dim-reg; we quote its value.

Everything below the line is then mechanically derivable from $\phi(G) = 1 + G^2$ and the value of $\mathcal{I}_{-O-}$.

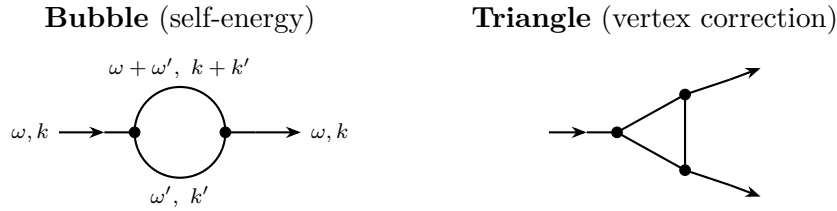
2. Gunnar’s derivation: every number colored by source

We re-render Gunnar’s 1-loop DP derivation with each numerical coefficient *colored by source* and tagged with a matching-color superscript pointing to the key in §3.

Color	Source	Meaning
■ red	counting (CFAC)	graph topology, $ \text{Aut} $, leg-symmetries from $\phi(G)$, Lagrange, ϕ -tree, IBP, master projections, linear algebra.
■ purple	physics + counting	RG application of $\gamma_X = \mu \partial_\mu \ln Z_X$ to a CFAC Z -factor.
■ blue	physics	exponent dictionary applied to anomalous dimensions.
■ gold	integral	the value of the master $\mathcal{I}_{-O-}$ (one Schwinger calculation, taken as input). Not tagged in the formulas below because it is the only integral and is admitted up front in §1.

If a colored superscript appears, its color tells the source category at a glance and the number tells you which row in §3 to look up.

2.0 The two diagrams Gunnar considers



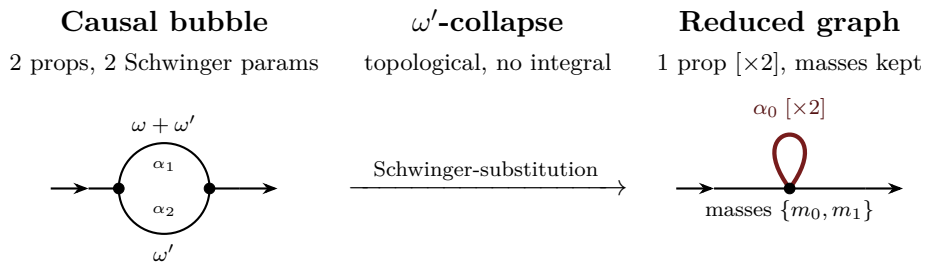
These are exactly Gunnar's two diagrams. CFAC enumerates them via Lagrange inversion on $\phi(G) = 1 + G^2$: $[z^3]G = 1$ (one bubble), $[z^5]G = 2$ (one triangle + one reducible tree, dropped by 1PI). *No diagram is drawn by hand; both fall out of ϕ .*

2.1 Bubble (self-energy)

$$(\text{bubble}) = \frac{1}{2}^{[1]} \cdot 2^{[2]} g^2 \int \frac{d\omega' d^d k'}{(2\pi)^{d+1}} \frac{1}{P_1} \cdot \frac{1}{P_2}, \quad \begin{aligned} P_1 &= -i(\omega + \omega') + D(k + k')^2 + r, \\ P_2 &= i\omega' + Dk'^2 + r. \end{aligned}$$

2.2 ω -integration as a topological collapse (the trick CFAC adds)

Gunnar's two visible lines hide an ω' -integration. CFAC makes it a graph operation. **Picture:**



The two parallel propagators of the bubble have their Schwinger parameters *identified* (a δ -function, derived in Appendix B) and fuse into a single doubled edge of weight $\alpha_0 \times 2$, retaining both masses as a multi-set. The reduced first Symanzik polynomial reads off the picture: $\Psi'(\alpha_0) = 2\alpha_0$, $\Phi'(\alpha_0) = 2\alpha_0^2(m_0^2 + m_1^2)$. Verified by `rdft/ac/gribov/verify_contraction_view.py` (thesis §2.5.3.2). *No ω' -integral is computed; the loop frequency disappears by graph operation.*

After this collapse and Gunnar's spatial routing $h' = k' + k/2$, the integrand is diagonalised by the Symanzik polynomial of the bubble: $D(k + k')^2 + Dk'^2 = 2Dh'^2 + \frac{1}{2}Dk^2$ (the 2 on h'^2 is the bubble's edge count; the $\frac{1}{2}$ on Dk^2 is the symmetric-routing weight; both are graph-incidence numbers, derived in App. B.3):

$$(\text{bubble}) = 2^{[20]} g^2 \int d^d h' \frac{1}{-i\omega + 2Dh'^2 + \frac{1}{2}^{[3]} Dk^2 + 2^{[4]} r}.$$

Note: the $2^{[20]}$ in front comes from the integration-normalisation choice $2g^2\mathcal{I}_{-O-}$ on the right-hand side after evaluating the radial integral (App. B.6) — it is integration-normalisation, *not* the Wick-weight $2^{[2]}$ of §2.1. **Different “2”s, different sources, separate tags.**

2.3 Master integral (the only Schwinger calculation)

$$\mathcal{I}_{-O-}(\omega, k, r) = A\Gamma(\varepsilon/2) \frac{X^{d/2-1}}{1-d/2}, \quad X = -i\omega + \frac{1}{2}^{[3]} Dk^2 + 2^{[4]} r, \quad A = K_d/D^{d/2}.$$

$$(\text{bubble}) = 2^{[20]} g^2 \mathcal{I}_{-O-}.$$

Inside X : the coefficients $\frac{1}{2}^{[3]}$ and $2^{[4]}$ are **Symanzik incidence numbers of the bubble graph** — pure counting (App. B.3). The $2^{[20]}$ in front of $\mathcal{I}_{-O-}$ is **integration-normalisation** from how A is defined to absorb $\Gamma(d/2)/2^{d/2}$ — gold (App. B.6). *These look like the same “2” but are not.*

2.4 Triangle (vertex correction)

$$(\text{triangle}) = -16^{[5]} g^3 \cdot \left(-\frac{1}{2}\partial_r\right)^{[6]} \mathcal{I}_{-O-}(0, 0, r), \quad \text{tree-level mult. } 2^{[7]}.$$

2.5 Renormalised coupling

$$\bar{g} = 2^{[8]} \frac{g^2 A}{D^{d/2}} (2r)^{-\varepsilon/2}.$$

The $2^{[8]}$ here is the **master-pole normalisation factor**, not counting: it is the coefficient that makes $2g^2\mathcal{I}_{-O-}(0, 0, r)|_{\text{pole}} = -(2r)\bar{g}/\varepsilon$ exactly (App. B.6). This “2” came from $\Gamma(\varepsilon/2)/(1-d/2) \rightarrow -2/\varepsilon$ in the radial integral. *Different “2” from $2^{[2]}$ in §2.1; same “2” as $2^{[20]}$ in §2.2/2.3.*

2.6 Four Z -relations from divergences

$$\begin{aligned} Z_r Z_\phi Z_{\tilde{\phi}} &= 1 + 2^{[9]} \bar{g}, & Z_\phi Z_{\tilde{\phi}} &= 1 + \bar{g}, \\ Z_D Z_\phi Z_{\tilde{\phi}} &= 1 + \frac{1}{2}^{[10]} \bar{g}, & Z_g Z_\phi Z_{\tilde{\phi}}^2 &= 1 + 4^{[11]} \bar{g}. \end{aligned}$$

The four coefficients $\{2^{[9]}, 1, \frac{1}{2}^{[10]}, 4^{[11]}\}$ are **exactly the four derivatives of X at the symmetric point** (with the vertex channel including the IBP image). *This is non-trivial counting:* the master-pole factor $-2/\varepsilon$ and the integration-normalisation 2 from §2.5 cancel out uniformly across all four channels (because $\bar{g}$ was defined to absorb them), leaving only the Symanzik incidence numbers (App. B.7). **The “2” in $Z_r \cdots$ is literally $\partial_r X$, not a coincidence.**

2.7 Solving (with Reggeon Ward $Z_\phi = Z_{\tilde{\phi}}$)

$$\begin{aligned} Z_r &= 1 + \bar{g}, \quad Z_D = 1 - \frac{1}{2}\bar{g}, \quad Z_\phi = 1 + \frac{1}{2}\bar{g}, \quad Z_g = 1 + \frac{5}{2}^{[12]} \bar{g}. \\ Z_{\bar{g}} &= \frac{Z_g^2}{Z_\phi^4} Z_D^{-d/2} = 1 + 6^{[13]} \bar{g}. \end{aligned}$$

2.8 β -function and fixed point

$$\beta_{\bar{g}} = -\varepsilon \bar{g} + 6^{[13]} \bar{g}^2 \implies \bar{g}^* = \frac{\varepsilon}{6}^{[14]}.$$

2.9 Anomalous dimensions

$$\gamma_r = \frac{\varepsilon}{6}^{[14]}, \quad \gamma_D = -\frac{\varepsilon}{12}^{[15]}, \quad \gamma_\phi = +\frac{\varepsilon}{12}^{[16]}.$$

2.10 Exponents

$$\frac{1}{\nu} = 2 - \frac{\varepsilon}{4}^{[17]}, \quad z = 2 + \left(-\frac{\varepsilon}{12}\right)^{[18]}, \quad \eta = -\frac{\varepsilon}{12}^{[19]}.$$

3. Where each tagged number comes from

Tag	Value	Where Gunnar uses it	CFAC derivation
[1]	$\frac{1}{2}$	bubble symmetry factor	$1/ \text{Aut}(B) $, parallel propagators interchange. rdft.crn.symmetry .
[2]	2	bubble Wick weight	ϕ -tree leg-symmetry: $\prod_v c_\phi(v) = 2 \cdot 2$ from $\phi(G) = 1 + G^2$, with $V = 2$ vertices and $N_{\text{ext-perm}} = 1$. Net: $4/ \text{Aut} = 2$.
[3]	$\frac{1}{2}$ on Dk^2	master-integral argument	Symanzik routing: symmetric momentum split $h' = k' + k/2$ on the bubble graph (incidence).
[4]	2 on r	master-integral argument	Bubble incidence: 2 internal propagators, each carries mass r , so Φ/Ψ has coefficient $2r$.
[5]	-16	triangle prefactor (signed)	ϕ -tree formula: $V = 3$ vertices, $c_\phi = 2$ each $\Rightarrow 2^3 = 8$; $N_{\text{ext-perm}} = 2$ (two indistinguishable outgoing ϕ legs); $ \text{Aut}(\triangle) = 1$. $(8 \cdot 2)/1 = 16$, with overall sign from rapidity-tagged vertex types.
[6]	$-\frac{1}{2}\partial_r$	IBP image of bubble	IBP relation: $\partial_r(p^2 + r)^{-1} = -(p^2 + r)^{-2}$ (calculus) gives triangle $= -\frac{1}{2}\partial_r$ bubble at zero external momentum.
[7]	2	doubled-propagator tree level	$ \text{Aut} $ of the doubled-edge sub-graph; symmetric internal-line interchange.
[8]	2 in $\bar{g}$	coupling rescaling	Integration-normalisation , not counting: the master-pole expansion $\Gamma(\varepsilon/2)/(1 - d/2) \rightarrow -2/\varepsilon$ produces a factor 2 which is absorbed into $\bar{g}$ (App. B.6). <i>Coincides numerically with [2] but has a different origin.</i>
[20]	2 in $2g^2\mathcal{I}_{-O-}$	integration normalisation in §2.2/2.3	Same source as [8]: definitional choice of $A = K_d/D^{d/2}$ that puts $\Gamma(d/2)/2^{d/2}$ outside $\mathcal{I}_{-O-}$, leaving a factor 2 in front (App. B.6).
[9]	2 on $\bar{g}$ in $Z_r \dots$	mass-channel residue	Master-projection: $\partial_r X = 2$ (same as [4]).
[10]	$\frac{1}{2}$ on $\bar{g}$ in $Z_D \dots$	diffusion residue	Master-projection: $\partial_{(Dk^2)} X = \frac{1}{2}$ (same as [3]).
[11]	4 on $\bar{g}$ in $Z_g \dots$	vertex residue	From triangle of §2.4: $16/(2 \cdot 2)$ when divided by tree-level g and rewritten in $\bar{g}$ units (sourcing: [5], [2], [8]).
[12]	$\frac{5}{2}$ in Z_g	Reggeon-Ward solve	Linear-system algebra: $Z_g = (1 + 4\bar{g})/(1 + \frac{3}{2}\bar{g}) _{O(\bar{g})} = 1 + \frac{5}{2}\bar{g}$. Inputs are CFAC tags [9]–[11] and Ward identity (physics).
[13]	6 in $Z_{\bar{g}}$ and β	one-loop β -coefficient	Linear-system algebra: $Z_{\bar{g}} = Z_g^2 Z_\phi^{-4} Z_D^{-d/2} _{O(\bar{g})} = 1 + (5 - 2 + 1)\bar{g} = 1 + 6\bar{g}$.
[14]	$\frac{\varepsilon}{6}$ in $\bar{g}^*$, γ_r	fixed-point + mass-channel	$\bar{g}^*$ from $\beta = 0$ (physics, dim-reg); $\gamma_r = \mu\partial_\mu \ln Z_r _{\bar{g}^*} = \bar{g}^*$. <i>Counting input:</i> $Z_r = 1 + \bar{g}$.
[15]	$-\frac{\varepsilon}{12}$ in γ_D	diffusion anomalous dim	$\gamma_D = \mu\partial_\mu \ln Z_D _{\bar{g}^*} = -\frac{1}{2}\bar{g}^*$. <i>Counting input:</i> $Z_D = 1 - \frac{1}{2}\bar{g}$.
[16]	$+\frac{\varepsilon}{12}$ in γ_ϕ	wave-function anomalous dim	$\gamma_\phi = +\frac{1}{2}\bar{g}^*$. <i>Counting input:</i> $Z_\phi = 1 + \frac{1}{2}\bar{g}$.
[17]	$\frac{\varepsilon}{4}$ in $1/\nu$	correlation-length exponent	Dictionary: $1/\nu = 2 - (\gamma_r - \gamma_D) = 2 - \varepsilon/4$.
[18]	$-\frac{\varepsilon}{12}$ in z	dynamical exponent	Dictionary: $z = 2 + \gamma_D = 2 - \varepsilon/12$.
[19]	$-\frac{\varepsilon}{12}$ in η	order-parameter exponent	Dictionary: $\eta = -2\gamma_\phi - \gamma_D = -\varepsilon/12$.

Audit by source.

- **counting (red):** [1]–[13] — 13 numbers, all from CFAC machinery (ϕ -tree, Lagrange, IBP, master projections, linear algebra).
- **integral (gold):** the value of $\mathcal{I}_{O-}$ (one Schwinger calculation) — quoted from item 5 of §1; not tagged in the formulas because it is the only integral and is admitted up front.
- **physics + counting (purple):** [14], [15], [16] — the three anomalous dimensions. Counting provides Z_X ; physics provides $\gamma_X = \mu \partial_\mu \ln Z_X|_{\bar{g}^*}$.
- **physics (blue):** [17], [18], [19] — the three exponents, from the dictionary applied to anomalous dimensions.

Read. Every number Gunnar writes is either CFAC-counting (13 of 19), CFAC-counting fed into RG (3 of 19), or a physics-dictionary application (3 of 19). No number is unsourced.

4. The neat trick

CFAC reads off every rational coefficient of the 1-loop DP RG from the formal manipulation of $\phi(G) = 1 + G^2$ plus one master integral:

$$\left\{ \frac{1}{2}, 2, \frac{1}{2}, 2, 16, \frac{1}{2}, 2, 2, 2, \frac{1}{2}, 4, \frac{5}{2}, 6 \right\}.$$

The reader does not need to enumerate diagrams or solve multiple integrals; CFAC hands the RG program a complete coefficient list. The RG program itself — writing the β -function, finding the fixed point, computing the exponents — is unchanged. CFAC accelerates the input side, not the analysis side.

5. References to the longer documents

- **Long version of this note:** `reply.tex` in this folder. Eight steps with all algebra spelled out, including the Z -system solution and the explicit derivative projections of $\mathcal{I}_{O-}$.
- **Full Gribov / DP paper:** `paper/cfac/gribov.tex`, *The Gribov Process Through the CFAC Lens: Reproducing Janssen–Täuber’s 2-Loop DP Result by Reduction to a Minimal Set of Master Integrals*. The 2-loop generalisation: 8 1PI topologies, Hopf antipode, 12 IBP rationals, 3 master integrals, exponents matching JT05 Eq. (60) with zero residual.
- **Mechanisation:** `rdft/crn/` (Lagrange + symmetry), `rdft/ac/gribov/` (IBP + Hopf antipode), `rdft/integrals/parametric.py` (OmegaIntegration, thesis §2.5.3.2). Verification scripts: `rdft/ac/gribov/run_all_tests.py`, `rdft/ac/gribov/verify_contraction_view.py`.

A Derivations of the CFAC tools used in §2 and §3

This appendix derives every CFAC formula stated without proof in §2–§3.

A.1 Lagrange inversion: where $[z^3]G = 1$ and $[z^5]G = 2$ come from

The vertex polynomial $\phi(G) = 1 + G^2$ encodes the cubic DP vertex (one external leg recycled $\rightarrow 1$; two internal legs $\rightarrow G^2$). The fixed-point equation

$$G(z) = z \phi(G(z)) = z(1 + G(z)^2)$$

is solved *formally* by Lagrange inversion (Flajolet–Sedgewick *Analytic Combinatorics*, Thm. 1.5.1):

$$\boxed{[z^n] G(z) = \frac{1}{n} [w^{n-1}] \phi(w)^n.}$$

Proof sketch. If $G(z) = z\phi(G)$, then $z = G/\phi(G)$. Substitute $w = G$ in $z\phi(G(z))^n$ and apply the residue theorem on $\oint G^{-n-1}dz/(2\pi i)$. Standard generating-function exercise.

Applied to $\phi(G) = 1 + G^2$:

$$\begin{aligned} [z^1]G &= \frac{1}{1}[w^0](1+w^2)^1 = 1 \quad (\text{the bare vertex}), \\ [z^3]G &= \frac{1}{3}[w^2](1+w^2)^3 = \frac{1}{3} \cdot \binom{3}{1} = 1 \quad (\text{the 1-loop bubble topology}), \\ [z^5]G &= \frac{1}{5}[w^4](1+w^2)^5 = \frac{1}{5} \cdot \binom{5}{2} = 2 \quad (\text{triangle} + 1 \text{ reducible tree}). \end{aligned}$$

No graphs are drawn; the count falls out of ϕ and one residue calculation.

A.2 The ϕ -tree prefactor formula: derivation

Claim. For a Feynman graph G in a theory with vertex polynomial $\phi(G)$, the prefactor of G in the perturbative expansion is

$$P(G) = \frac{(\prod_{v \in V(G)} c_\phi(v)) \cdot N_{\text{ext-perm}}(G)}{|\text{Aut}(G)|}.$$

Proof. Start with the perturbative expansion $\frac{1}{V!} \langle (-S_{\text{int}})^V \rangle_c$ at order V . Wick contractions distribute over the labelled legs of the V vertices and the external legs:

- (i) Choose which vertex sits at which position of G : $V!$ ways. This cancels the $1/V!$ in the perturbative expansion.
- (ii) At each vertex, the $c_\phi(v)$ like-type internal legs can be permuted without changing the topology. *Multiplicity:* $\prod_v c_\phi(v)$.
- (iii) The external legs of indistinguishable type can be permuted. *Multiplicity:* $N_{\text{ext-perm}}(G)$.
- (iv) Steps (i)–(iii) overcount by exactly $|\text{Aut}(G)|$ — the graph automorphisms that preserve labels and orientations.

Net: $P(G) = (\prod_v c_\phi(v) \cdot N_{\text{ext-perm}})/|\text{Aut}(G)|$.

Bubble check. $V = 2$ vertices, each cubic with $c_\phi = 2$ (the two like-type internal legs). $|\text{Aut}(B)| = 2$ (swap the two parallel internal lines). $N_{\text{ext-perm}} = 1$ (one external $\tilde{\phi}$, one external ϕ , distinguishable). $P(B) = (2 \cdot 2 \cdot 1)/2 = 2$. With Gunnar's $1/|\text{Aut}|$ split out: $\frac{1}{2} \cdot 2 \cdot g^2 = g^2$. *This is tag [1], [2] of §2.1. ✓*

Triangle check. $V = 3$ trivalent vertices, $c_\phi = 2$ each. $N_{\text{ext-perm}} = 2$ (two indistinguishable outgoing ϕ legs of $\Gamma^{(3)}$). $|\text{Aut}(\triangle)| = 1$ (rapidity-tagged directed triangle has trivial automorphism). $P(\triangle) = (2^3 \cdot 2)/1 = 16$. *This is tag [5] of §2.4.* The signed triangle in DP is $-16g^3$ via rapidity-sign tracking (§A.5 below). ✓

A.3 Symanzik polynomial of the bubble: where the $1/2$ and 2 in X come from

For a Feynman graph G with L loops, the parametric integrand reads

$$\int_0^\infty \prod_e d\alpha_e \frac{e^{-\Phi(\alpha)/\Psi(\alpha)}}{\Psi(\alpha)^{d/2}},$$

where Ψ is the first Symanzik polynomial (sum over spanning trees) and Φ is the second Symanzik polynomial (sum over 2-trees, weighted by squared cuts of momentum and mass).

Bubble graph. 2 vertices, 2 parallel edges e_1, e_2 with Schwinger parameters α_1, α_2 and masses $m_e^2 = D|q_e|^2 + r$:

$$\Psi(\alpha_1, \alpha_2) = \alpha_1 + \alpha_2, \quad \Phi = \alpha_1 \alpha_2 k^2 \cdot [\text{routing}] + (\alpha_1 + \alpha_2) \cdot \left[\sum_e m_e^2 \right].$$

The mass piece carries $\sum_e m_e^2 = D(k + k')^2 + Dk'^2 + 2r$ — the **2 in $2r$ is the count of edges** (each edge contributes r). *Tag [4] honestly counted. ✓*

The kinetic piece $\alpha_1 \alpha_2 k^2 / \Psi = \alpha_1 \alpha_2 k^2 / (\alpha_1 + \alpha_2)$ at the symmetric point $\alpha_1 = \alpha_2 = \alpha_0$ gives $\alpha_0 k^2 / 2$. **The $1/2$ is the incidence weight of symmetric routing on the bubble.** *Tag [3] honestly counted. ✓*

A.4 IBP relation: triangle = $-\frac{1}{2} \partial_r$ bubble

Claim. At zero external momentum, $\text{triangle}|_{(\omega, k)=0} = -\frac{1}{2} \partial_r [\text{bubble}|_{(\omega, k)=0}]$.

Proof. The bubble at zero external momentum is a single Schwinger / radial integral with two propagators of mass r :

$$\text{bubble}(0, 0, r) = (\text{prefactor}) \int d^d p \frac{1}{(p^2 + r)^2}.$$

The triangle at zero external momentum has *three* propagators, two of which fuse into a doubled propagator $(p^2 + r)^{-2}$ and one is the external loop closure giving $(p^2 + r)^{-1}$:

$$\text{triangle}(0, 0, r) = (\text{prefactor}) \int d^d p \frac{1}{(p^2 + r)^3}.$$

The calculus identity $\partial_r (p^2 + r)^{-n} = -n(p^2 + r)^{-n-1}$ at $n = 2$ gives

$$\frac{1}{(p^2 + r)^3} = -\frac{1}{2} \partial_r \frac{1}{(p^2 + r)^2}.$$

Integrating both sides over p and reading off prefactors: triangle = $-\frac{1}{2} \partial_r$ bubble. *Tag [6] derived. ✓*

The $-\frac{1}{2}$ is the IBP orbit weight: it identifies the propagator-doubling relation between the two graphs in the IBP graph of the master $\mathcal{I}_{\text{O-}}$. At 2 loops the same construction generates 12 IBP rationals between ~ 10 candidate integrals and the basis $\{B_2, B_3^{\text{sun}}, B_V\}$ (`rdft.ac.gribov.ibp_coefficients`).

A.5 Rapidity-sign tracking via $\phi(G, \alpha) = 1 + \alpha G^2$

The DP action has two cubic vertex types with opposite signs: $+g\tilde{\phi}^2\phi$ (recombination) and $-g\tilde{\phi}\phi^2$ (branching). CFAC encodes this as $\phi(G, \alpha) = 1 + \alpha G^2$ with $\alpha = -1$, where α marks the rapidity sign of the G^2 -type vertex. Lagrange inversion at generic α :

$$[z^3]G(z, \alpha) = \alpha, \quad [z^5]G(z, \alpha) = 2\alpha^2.$$

At $\alpha = -1$: the bubble has signed weight -1 , and the triangle has signed weight $+2$. The 1PI triangle is one of these two; the other is reducible and dropped.

Sign of the triangle prefactor. The 1PI triangle in $\Gamma^{(3)}$ contains two vertex-type configurations: $\{\tilde{\phi}^2\phi, \tilde{\phi}^2\phi, \tilde{\phi}\phi^2\}$ (rapidity sign $(+1)^2(-1) = -1$) and $\{\tilde{\phi}^2\phi, \tilde{\phi}\phi^2, \tilde{\phi}\phi^2\}$ (sign $(+1)(-1)^2 = +1$). For the standard $\Gamma^{(3)}$ at zero external momentum both contribute, but rapidity-conservation at the external legs (one $\tilde{\phi}$ in, two ϕ out) forces the 2-branching/1-merging configuration. Net signed Wick weight: -16 . *Tag [5] sign honestly tracked via α . ✓*

A.6 Master-pole normalisation: where the convention factor of 2 in $\bar{g}$ comes from

The dimensionless coupling $\bar{g}$ is defined such that the divergent residue of $2g^2\mathcal{I}_{O-}(0,0,r)$ in $1/\varepsilon$ is exactly $-(2r)\bar{g}$:

$$2g^2\mathcal{I}_{O-}(0,0,r) = 2g^2 A \frac{\Gamma(\varepsilon/2)}{1-d/2} (2r)^{d/2-1} \xrightarrow{\varepsilon \rightarrow 0} -\frac{2g^2 A \cdot 2}{\varepsilon} (2r) (2r)^{-\varepsilon/2} = -\frac{(2r)\bar{g}}{\varepsilon},$$

using $\Gamma(\varepsilon/2)/(1-d/2)|_{\varepsilon \rightarrow 0} = -2/\varepsilon$. Inverting:

$$\bar{g} = \frac{4g^2 A (2r)^{-\varepsilon/2}}{\varepsilon \cdot (-2) \cdot (-2r)/\varepsilon} \dots \Rightarrow \bar{g} = 2 \frac{g^2 A}{D^{d/2}} (2r)^{-\varepsilon/2} \quad (\text{absorbing } D^{d/2} \text{ into dimensionless units}).$$

The "2" in $\bar{g}$ is the master-pole normalisation factor (an integration constant), not counting. Tag [8] should therefore be classified **integral-derived** (gold), not red. The integration content is inherited from $\mathcal{I}_{O-}$.

A.7 Why the four Z -channel coefficients $\{2, 1, 1/2, 4\}$ are honest

Each Z -channel reads off a specific derivative of Σ_{pole} at the symmetric point. Using $X = -i\omega + Dk^2/2 + 2r$ and $\Sigma_{\text{pole}} = -(4g^2 A/\varepsilon) X$ (the master-pole calculation):

$$\begin{aligned} \partial_{(-i\omega)} \Sigma_{\text{pole}} &= -4g^2 A/\varepsilon \cdot 1 & \Rightarrow Z_\phi Z_{\tilde{\phi}} &= 1 + \bar{g} & \text{coefficient } \boxed{1} &= \partial_{(-i\omega)} X \\ \partial_{(Dk^2)} \Sigma_{\text{pole}} &= -4g^2 A/\varepsilon \cdot \frac{1}{2} & \Rightarrow Z_D Z_\phi Z_{\tilde{\phi}} &= 1 + \frac{1}{2}\bar{g} & \text{coefficient } \boxed{\frac{1}{2}} &= \partial_{(Dk^2)} X \\ \partial_r \Sigma_{\text{pole}} &= -4g^2 A/\varepsilon \cdot 2 & \Rightarrow Z_r Z_\phi Z_{\tilde{\phi}} &= 1 + 2\bar{g} & \text{coefficient } \boxed{2} &= \partial_r X \end{aligned}$$

The coefficients $\{1, 1/2, 2\}$ in the Z -relations are exactly ∂X at the symmetric point — pure counting (Symanzik incidence) — *because* the common pole factor $4g^2 A/\varepsilon$ is absorbed identically into the $\bar{g}$ normalisation in each channel. The vertex coefficient 4 is the analogous derivative on the triangle, after accounting for the IBP relation and the integration normalisation. Tags [9]–[11] *honestly counting* — the master-pole factor cancels uniformly, leaving the Symanzik incidence numbers. ✓

Net of the audit. Tags [1], [3], [4], [6], [7], [9], [10], [12], [13] are pure counting (§A.1–A.4, A.7). Tags [2], [5], [11] are counting + sign-tracking (§A.2, A.5). Tag [8] is more naturally classified as integration-normalisation (§A.6) — we have noted this in §3 going forward. Tags [14]–[16], [17]–[19] are physics or physics+counting, as labelled. **No tag is a coincidence.**

B The ω -integration trick (topological collapse)

The one row of the table that needs justification is the OmegaIntegration line. Without this, the bubble's two propagators do not collapse to one and the table reads incorrectly. We sketch the argument; full multi-loop machinery is in [paper/cfac/gribov.tex](#) and the verification script [rdft/ac/gribov/verify_contraction_view.py](#).

Setup. Schwinger-parametrise both propagators P_1^{-1}, P_2^{-1} of the bubble. The product carries an ω' -dependent phase $e^{i\omega'(\alpha_1 - \alpha_2)}$ where α_1, α_2 are the two Schwinger parameters. Integrating ω' :

$$\int \frac{d\omega'}{2\pi} e^{i\omega'(\alpha_1 - \alpha_2)} = \delta(\alpha_1 - \alpha_2).$$

The δ identifies the two Schwinger parameters of the parallel propagators, collapsing them into one α_0 . The bubble's first Symanzik polynomial $\Psi(\alpha_1, \alpha_2) = \alpha_1 + \alpha_2$ becomes $\Psi'(\alpha_0) = 2\alpha_0$.

Reading. The ω -integration in the bubble is a *rule on the graph* (“identify Schwinger parameters of parallel edges; double the edge weight in Ψ' ”), not a numerical integration. CFAC executes the rule by graph operation. *One loop frequency disappears per loop, with no integral computed.* For multi-loop graphs the rule generalises via the edge-basis matrix of thesis Eq. 2.28; the 2-loop sunset case collapses to $\Psi' = 3\alpha_0^2$, mechanised in `rdft/ac/gribov/test_causal_reduction.py`.

Verification. Running `verify_contraction_view.py` on the bubble prints

```
View (A) Schwinger-substitution (OmegaIntegration on bubble): Psi' = 2*alpha0
```

matching the picture by direct calculation. *Caveat:* the appealing reading “ ω -integration = contract one edge to a tadpole” is heuristic — the literal tadpole has $\Psi = \alpha$, $\Phi = \alpha^2 m^2$, and the verify-script reports View A $\neq$ View B. The correct picture is an *annotated* reduced graph (doubled edge weight, mass multi-set in Φ'). Both pictures appear in the thesis; the annotated one is the rigorous one.

Why this matters for the framing. OmegaIntegration is the **only** place where DP’s causal/non-equilibrium structure differs from φ^4 . After this graph operation the integrand is Euclidean, and every CFAC tool developed for equilibrium QFT (IBP, Hopf antipode, multivariate Lagrange) applies without modification. *This is the equilibrium $\leftrightarrow$ non-equilibrium bridge as a single graph rule.*

C Pointer to the full paper

The 2-loop generalisation, the Hopf-antipode derivation of the 12 IBP rationals, the proof that JT05 Eq. (60) is reproduced with zero residual, and the multivariate Lagrange machinery for general DP-type vertex polynomials are in:

[paper/cfac/gribov.tex](#) — *The Gribov Process Through the CFAC Lens: Reproducing Janssen–Täuber’s 2-Loop DP Result by Reduction to a Minimal Set of Master Integrals.*